

Job Search AI Agent Development Project – Dual Track Version

Team Size: 3–4 students | **Duration:** 8 Weeks | **Choose Your Challenge Level**

Project Overview

Build an AI-powered career assistant that automates job searching, optimizes resumes for ATS systems, provides company insights, and conducts mock interviews. You'll master job market APIs, resume parsing, and interview automation while creating a comprehensive career development platform.

Domain Focus: Career Development & Job Search Automation
Core Skills: Job market analysis, resume optimization, interview preparation, career guidance

CHOOSE YOUR TRACK

Track A: Essential (Recommended for Most Teams)

Focus: Core job search skills, achievable goals, strong fundamentals

- **Domain:** Job Search Assistant OR Resume Optimizer
- **APIs:** 2–3 basic integrations (Job search + Company data + Resume parsing)
- **Framework:** Standard LangChain with career tools
- **UI:** Streamlit with job listings display
- **Deployment:** Single platform (Streamlit Cloud)
- **Features:** Job matching, basic resume feedback, company research

Track B: Advanced (For Confident Teams)

Focus: Industry-ready HR-tech skills, complex career workflows

- **Domain:** Comprehensive Career Development Platform
- **APIs:** 5+ integrations with advanced error handling
- **Framework:** LangGraph for complex career workflows
- **UI:** Professional React/Next.js with applicant tracking
- **Deployment:** Multi-service architecture with user authentication

- **Features:** ATS optimization, interview scheduling, salary negotiation, career path planning

Learning Outcomes

By completing this project, you will:

- **Job Market Intelligence:** Integrate with major job platforms and analyze market trends
- **Resume Engineering:** Build ATS-optimized resume parsing and enhancement systems
- **Interview Automation:** Create AI-powered mock interview systems with feedback
- **Career Analytics:** Develop insights on job market trends and salary benchmarking
- **User Experience:** Design intuitive interfaces for job seekers and career changers
- **Professional Networking:** Understand recruiter workflows and applicant tracking systems

Indian Job Market APIs & Data Sources

Primary Indian Job Platforms:

- **Naukri.com API:** India's largest job portal
- **Monster India:** Job listings and company data
- **TimesJobs API:** Times group job platform
- **Indeed India:** Global platform with Indian jobs
- **LinkedIn Jobs API:** Professional network integration
- **Shine.com:** Indian job search platform
- **Freshersworld:** Entry-level and fresher jobs

Company Intelligence (India-focused):

- **Glassdoor API:** Company reviews and salary data
- **AmbitionBox:** Indian company reviews and culture insights
- **Cutshort:** Startup jobs and company profiles
- **Angellist:** Startup ecosystem and funding data
- **Crunchbase:** Company funding and growth data

Resume & Skills APIs:

- **Resume parsing:** Sovren, Affinda, or open-source alternatives
- **Skills assessment:** HackerRank API, Codility
- **LinkedIn Profile API:** Professional profile extraction
- **GitHub API:** Developer portfolio analysis

Indian Specific Features:

- **Notice Period handling:** 30/60/90 days common in India
- **Salary in LPA:** Lakhs per annum format
- **Location preferences:** Metro vs Tier-2 cities
- **Work from home:** Post-COVID preferences

Week-by-Week Structure

Week 1-2: Foundation & Quick Win

Goal: Get basic job search functionality working and deployed

Track A Checklist:

- [] Create GitHub repo with career project structure
- [] Set up development environment (Python, requests, BeautifulSoup, LangChain)
- [] Build basic job search chatbot with Indian job sites
- [] Implement simple job listing scraper (Indeed/Naukri with rate limiting)
- [] Add job filtering by location, experience, salary
- [] Deploy basic version on Streamlit Cloud
- [] Record 2-minute demo showing job search queries
- [] **Milestone:** Working deployed demo with Indian job listings

Track B Checklist:

- [] All Track A requirements plus:
- [] Set up advanced environment (LangGraph, PostgreSQL, Redis, FastAPI)
- [] Implement user authentication and profile management
- [] Add comprehensive error handling for API failures and rate limits
- [] Set up monitoring for job search performance and user engagement
- [] Create CI/CD pipeline with automated testing

Week 3-4: Core Career Agent Architecture

Goal: Transform into a multi-functional career assistant

Track A Checklist:

- ☐ Integrate 2 career tools (job search + company research)
- ☐ Implement basic resume parsing (PDF to text extraction)
- ☐ Add simple job matching based on skills and preferences
- ☐ Create company information lookup functionality
- ☐ Handle multiple job platforms with unified search
- ☐ **Milestone:** Agent provides comprehensive job search assistance

Track B Checklist:

- ☐ All Track A requirements plus:
- ☐ Implement complex agent workflow with LangGraph for career planning
- ☐ Add 4+ career tools (jobs, resume, interview prep, salary research, networking)
- ☐ Implement sophisticated resume analysis using NLP models
- ☐ Add application tracking system integration
- ☐ Create comprehensive user profile with career goals and preferences
- ☐ Implement job alert system with email/SMS notifications

Week 5-6: Domain Specialization

Track A Options (Choose ONE):

Option A1: Indian Job Search Specialist

- ☐ Integrate Naukri.com and TimesJobs APIs
- ☐ Add SQLite database for saving job applications and preferences
- ☐ Create location-based job search (Mumbai, Bangalore, Delhi, Pune, Hyderabad)
- ☐ Implement salary conversion and comparison (LPA format)
- ☐ Add notice period and work-from-home filtering
- ☐ Handle Indian company culture and benefits analysis

Option A2: Resume Optimization Engine

- ☐ Create ATS-friendly resume analysis and scoring
- ☐ Add keyword optimization based on job descriptions
- ☐ Implement skills gap analysis with learning recommendations
- ☐ Generate tailored resume versions for different job applications
- ☐ Add LinkedIn profile optimization suggestions

Track B Requirements:

- [] Integrate 5+ job and career APIs with sophisticated fallback strategies
- [] Design complex database schema for user profiles, job applications, and career tracking
- [] Implement advanced interview preparation with AI-powered mock interviews
- [] Add real-time salary benchmarking and negotiation assistance
- [] Create career path recommendations with skill development roadmaps
- [] **Advanced:** Implement calendar integration for interview scheduling and follow-ups

Week 7–8: Polish & Production

Track A Checklist:

- [] Create professional Streamlit interface with:
 - [] Job search dashboard with filters and sorting
 - [] Resume upload and analysis functionality
 - [] Company research and insights display
 - [] Job application tracking cards
 - [] Export functionality (PDF job lists, resume versions)
- [] Add input validation for resume uploads and search parameters
- [] Implement proper error messages for API failures
- [] Write comprehensive README with setup instructions and API configurations
- [] Prepare final demo video (5–7 minutes) showcasing Indian job market features
- [] **Milestone:** Production-ready career assistance application

Track B Checklist:

- [] Build professional React/Next.js interface with advanced UX/UI
- [] Deploy with proper microservices architecture (auth, jobs, resume, interview services)
- [] Add comprehensive analytics dashboard for job search performance
- [] Implement career data security best practices (resume privacy, application tracking)
- [] Add performance optimization for real-time job searching and notifications
- [] Create technical documentation including API documentation and user guides
- [] Prepare comprehensive demo (8–10 minutes) with end-to-end job search workflow

Assessment Criteria

Track A Grading (100 points):

- **Career Functionality (40 points):** Working agent with 2+ career tools + database
- **Job Matching Accuracy (20 points):** Relevant job recommendations and filtering
- **User Interface (15 points):** Clean, functional Streamlit app with intuitive navigation
- **Deployment (15 points):** Stable hosted application with live job data
- **Documentation (10 points):** Clear README with career setup instructions

Track B Grading (120 points + bonuses):

- **Advanced Career Architecture (45 points):** Complex workflows, interview automation
- **Professional HR-Tech UI/UX (25 points):** Advanced interface, user experience design
- **Production Deployment (20 points):** Multi-service architecture with authentication
- **Career Code Quality (15 points):** Testing, security, data privacy compliance
- **Technical Presentation (15 points):** Career system architecture, algorithm explanation
- **Bonus Features (up to 20 points):** AI interviews, salary negotiation, career coaching

Weekly Checkpoint System

Mandatory Checkpoints (Both Tracks):

- **Week 2:** Deployed working demo with Indian job search functionality
- **Week 4:** Agent with multiple career tools functioning
- **Week 6:** Specialized career features integrated and tested
- **Week 8:** Final submission with comprehensive career assistance

Git & Collaboration Requirements:

- [] Regular commits with proper data handling (never commit API keys or user data!)
- [] Proper branching strategy for feature development
- [] Code reviews via pull requests with focus on user privacy

- [] Issues tracking for career features and user experience bugs
- [] Environment variables for all API keys and sensitive configurations
- [] Tagged final release (v1.0) with production-ready deployment

Suggested Tech Stacks

Track A Stack:

- **Frontend:** Streamlit with custom components for job displays
- **Backend:** Python + LangChain + requests + BeautifulSoup
- **Database:** SQLite for user preferences and job tracking
- **LLM:** Gemini (free tier) or GPT-3.5-turbo
- **Deployment:** Streamlit Cloud
- **Career APIs:** Indeed, Glassdoor (free tiers), web scraping with ethical limits

Track B Stack:

- **Frontend:** React/Next.js + Tailwind CSS + Material-UI
- **Backend:** FastAPI + LangGraph + Celery (async job processing)
- **Database:** PostgreSQL + Redis (job caching and user sessions)
- **LLM:** GPT-4 or Claude for sophisticated career advice
- **Deployment:** Vercel (frontend) + Railway/Render (backend)
- **Authentication:** Auth0 or Firebase Auth
- **Monitoring:** Custom career analytics dashboard

Success Metrics

Track A Success:

- 90%+ teams deploy working career application with Indian job market support
- 80%+ teams demonstrate accurate job matching and resume analysis
- All teams can explain career development workflows and API integrations
- Portfolio-worthy HR-tech project for internship applications

Track B Success:

- Production-grade career platform with real-time job search capabilities
- Industry-standard user data security and privacy compliance
- Advanced career coaching and interview preparation implementation

- Interview-ready technical presentation for HR-tech and career services roles

Submission Requirements

All Teams Submit:

GitHub Repository with:

- Complete source code with career data processing and privacy protection
- README with setup instructions and API key configuration
- Demo video (5-10 minutes) showcasing comprehensive career assistance
- Architecture documentation explaining job search algorithms and data flow
- Live deployment with working URL and sample user journeys

Viva Presentation covering:

- Career API integration strategies and ethical data collection
- Technical architecture decisions for handling user data and privacy
- Job matching algorithms and resume optimization techniques
- Individual team member contributions and learning outcomes
- Future improvement ideas for enhanced career features

Track B Additional Requirements:

- Technical blog post about HR-tech development and user privacy challenges
- Performance benchmarks for job search speed and accuracy
- Privacy and security assessment documentation for user data handling
- API documentation for career services and integration guidelines

Recognition System

Achievement Badges:

- **Pioneer** 🚀 : First team to deploy (Week 2)
- **Master** 🎯 : Most creative tool integrations
- **UX Champion** 🎨 : Best user interface design
- **Security First** 🛡️ : Best security practices
- **Data Driven** 📊 : Best monitoring and analytics
- **Innovation Award** ✨ : Most creative overall approach

Portfolio Benefits:

- **LinkedIn Showcase:** Help teams create professional HR-tech project posts
- **GitHub Stars:** Instructor promotion of best career development repositories
- **Industry Connections:** Introduction to Indian HR-tech companies and recruitment firms
- **Internship Opportunities:** Showcase to companies like Naukri, Monster, Indeed, and HR startups

Frequently Asked Questions

Q: How do I handle job search without violating website terms of service?

A: Use official APIs when available, implement respectful rate limiting for scraping, and always check robots.txt and terms of service.

Q: What about user privacy and resume data security?

A: Never store sensitive user data without consent, implement proper encryption, and provide data deletion options. Consider GDPR compliance.

Q: How do I make job matching accurate for Indian job market?

A: Understand Indian work culture, notice periods, salary structures (LPA), and location preferences. Train on Indian job descriptions.

Q: Can we switch tracks mid-project?

A: Yes, from B to A after Week 4 checkpoint. Consider complexity of user authentication and data handling when deciding.

Q: How do I handle different resume formats and ATS systems?

A: Focus on common formats (PDF, DOCX), implement robust parsing, and test with actual ATS systems if possible.

Q: What about mock interview implementation?

A: Track A: Text-based Q&A with feedback. Track B: Voice/video integration with speech analysis and real-time responses.

Q: How do I benchmark salary data for Indian market?

A: Use Glassdoor, PayScale, and AmbitionBox APIs. Consider experience level, location (metro vs tier-2), and industry variations.

Q: What ethical considerations should I keep in mind?

A: Respect user privacy, avoid bias in job matching, provide honest career advice, and maintain transparency about AI limitations in career guidance.